
Report on Discriminatory PointNet & PointNet++ Models

Summary

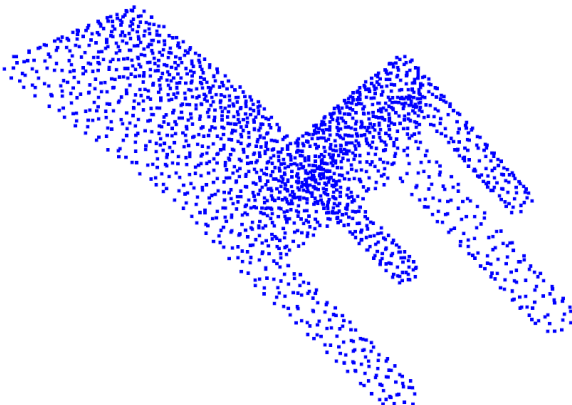
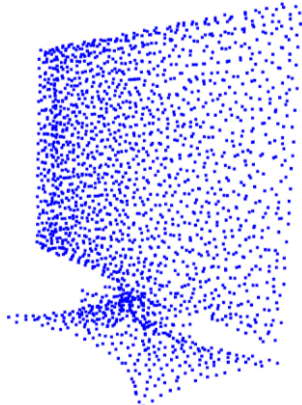
1. Overview of datasets used and processing techniques.
2. Exploration on discriminatory PointNet and PointNet++ model that are used in generative point cloud model. Results and comparison to literature.
3. Next steps.

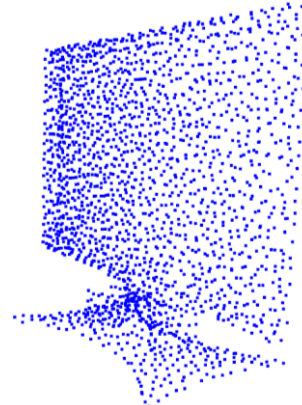
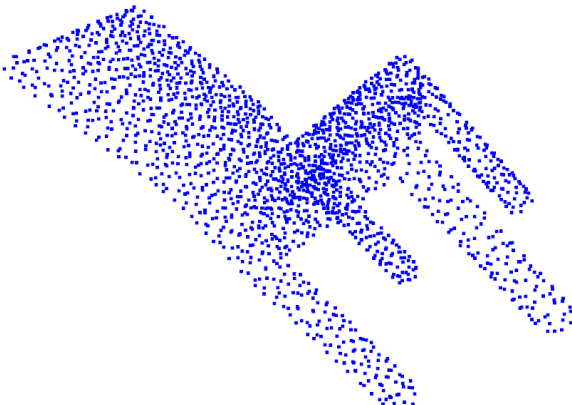
1 Datasets

All preprocessing steps are detailed in the README file from the [GitHub](#) repository.

ModelNet [1]

The dataset used is the **ModelNet40_Normal_Resampled**, the dataset features 40 classes with 9843 instances for training and 2468 instances for testing from ModelNet40. All models have normal annotations and are resampled to have uniform point density. Normal annotations can be dropped during dataset loading to only train the model on three coordinates (x, y, z). 10 classes can also be selected out to form ModelNet10. Data can be downloaded from [Kaggle](#). The original datasets and aligned version can be downloaded from [official website](#).

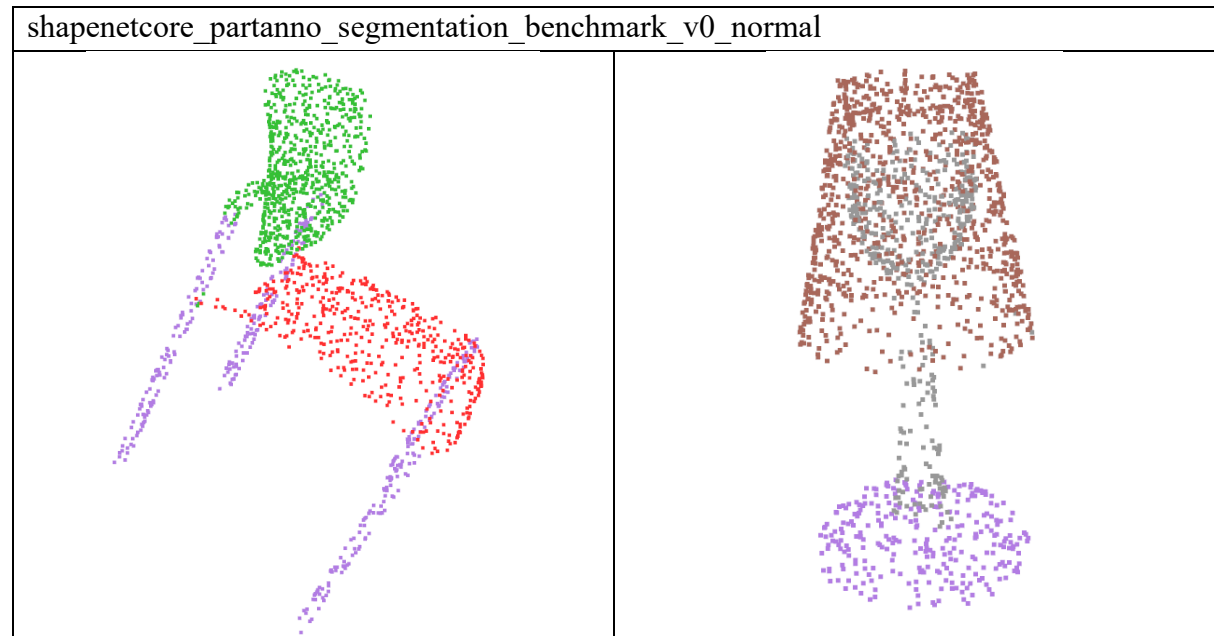
ModelNet40_Normal_Resampled	
	



During the first run, it will take longer to preprocess data (around 60 minutes for both train and test at 2048 points), the generated .dat files will be used in subsequent runs to reduce overhead from processing.

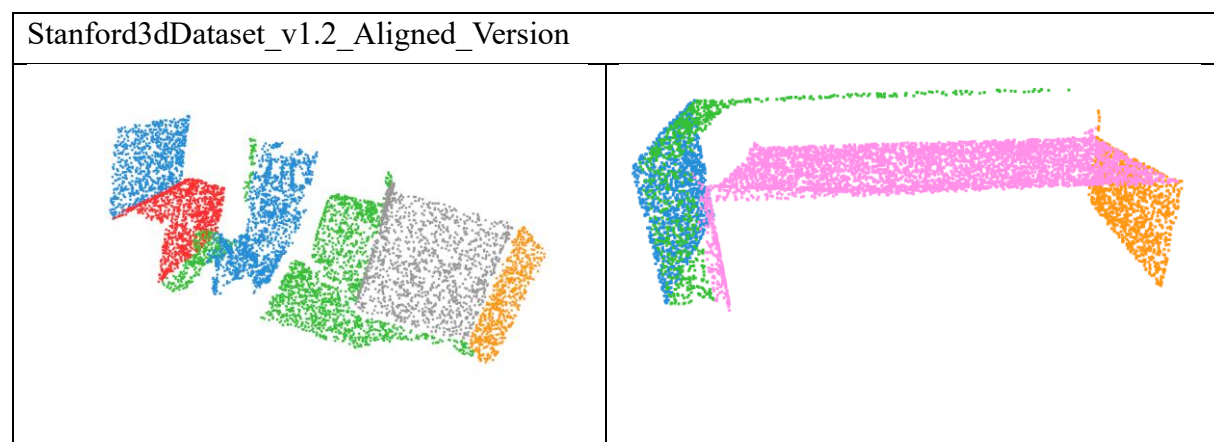
ShapeNetCore [2]

The dataset used is the **shapenetcore_partanno_segmentation_benchmark_v0_normal**, the dataset features 16 classes with 12137 instances for training and 2874 instances for testing. All models have normal annotations. Normal annotations can be dropped during dataset loading to only train the model on three coordinates (x, y, z). Data can be downloaded from [Kaggle](#).



Stanford Large-Scale Indoor Spaces 3D Dataset (S3DIS) [3]

The dataset used is **Stanford3dDataset_v1.2_Aligned_Version**, the dataset consists of five large-scale indoor areas from three different buildings, each covering approximately 1900, 450, 1700, 870 and 1100 square meters (total of 6020 square meters). The areas have diverse architectural style and appearance, they include mainly office areas, educational and exhibition spaces, and conference rooms, personal offices, restrooms, open spaces, lobbies, stairways, and hallways, etc. Data can be downloaded after filling up the [Google Form](#).



More samples are at [appendix \(j\)](#).

2 PointNet & PointNet++

Results

The PointNet [4] and PointNet++ MSG (multi-scale grouping) [5] are trained for 50 epoch only with 2048 points and batch size of 32. Plot of test accuracy against time at [appendix \(a\)](#) and [appendix \(b\)](#). Trained model weights at [appendix \(c\)](#).

a) Classification on ModelNet10, logs at [appendix \(d\)](#)

Model	Details	Accuracy (%)
PointNet	PyTorch, only xyz	92.78
	PyTorch, with normal	94.18
PointNet++ (MSG)	PyTorch, only xyz	93.68
	PyTorch, with normal	94.61

b) Classification on ModelNet40, logs at [appendix \(d\)](#)

Model	Details	Accuracy (%)
PointNet	Paper	89.20
	PyTorch, only xyz	88.94
	PyTorch, with normal	90.38
PointNet++ (MSG)	Paper	90.70
	PyTorch, only xyz	92.07
	PyTorch, with normal	92.34

c) Part Segmentation on ShapeNetCore, logs at [appendix \(e\)](#)

Model	Details	Instance Avg IoU	Class Avg IoU
PointNet	Paper	83.70	80.40
	PyTorch, only xyz	83.26	78.25
	PyTorch, with normal	83.93	80.46
PointNet++ (MSG)	Paper	85.10	81.90
	PyTorch, only xyz	84.79	81.14
	PyTorch, with normal	85.13	82.16

d) Semantic Segmentation on S3DIS, logs at [appendix \(f\)](#).

Model	Details	Overall Acc	Class Avg IoU
PointNet	Paper	78.62	47.71
	PyTorch	79.79	43.03
PointNet++ (MSG)	Paper	N/A	N/A
	PyTorch	82.57	55.57

Sample outputs

Sample outputs can be found at [appendix \(g\)](#), [appendix \(h\)](#) and [appendix \(i\)](#).

Conclusion

Learning and exploration of the discriminative point cloud models are important because generative point cloud models are derived from [6] or contain encoder module that make use of discriminative model layers [7,8]. This is due to the properties 3D point cloud data – unordered, interaction among points, and invariance under transformation [1]. Therefore, a discriminative model that can reliably extract features from 3D point cloud data is crucial for the success of generating accurate and detailed point cloud data. Below is an example of semantic segmentation output from PointNet and PointNet++. It can be seen that PointNet++ can more reliably capture details of the scene and output correct semantic segmentation labels.

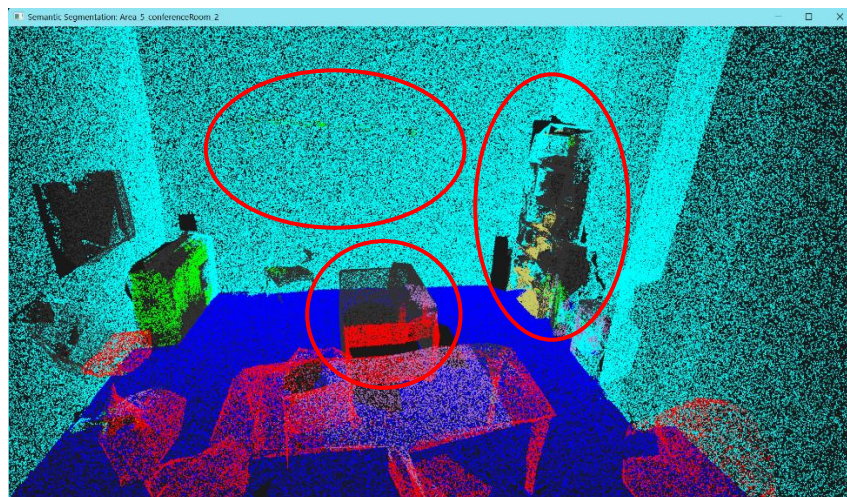


Figure 1: Semantic segmentation by PointNet of test Area 5 Conference Room 2

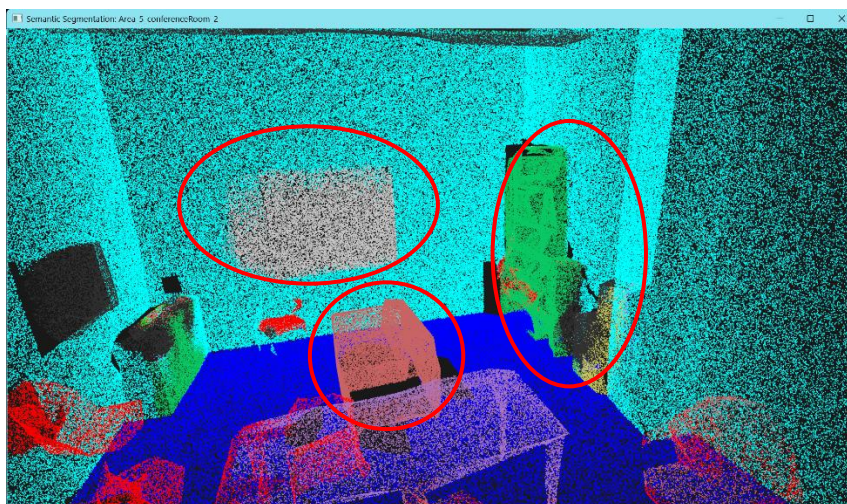


Figure 2: Semantic Segmentation by PointNet++ of test Area 5 Conference Room 2

This difference is due to the new techniques used in PointNet++, multi-scale grouping (MSG). By performing sampling (using farthest point sampling), grouping (of points within a Euclidean space of the point) and applying PointNet layer repetitively, it can progressively abstract larger and larger local regions. This improvement gained by applying different model architecture motivates further exploration into newer discriminative point cloud model architectures to form the generative point cloud model.

3 Next Steps

- Learning and experimenting with Generative PointNet [6].
- Obtaining and preprocessing scene (rural terrain and electrical transmission system) [12], indoor space [10,11], structures (exterior architectural structures) [13] datasets. List of 177 3D point cloud datasets for industry-specific applications [14].
- Exploring transformers in 3D point cloud models [9].

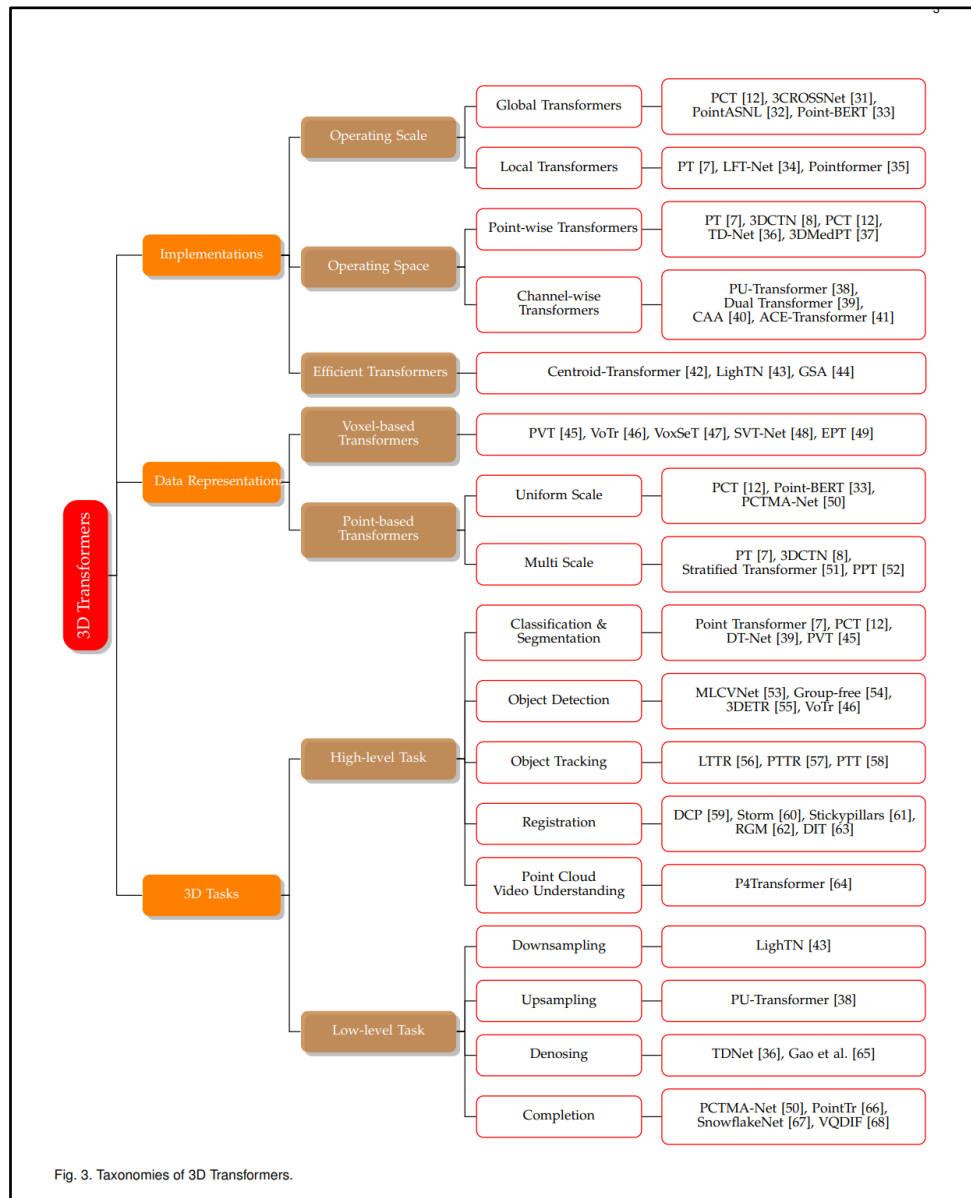


Fig. 3. Taxonomies of 3D Transformers.

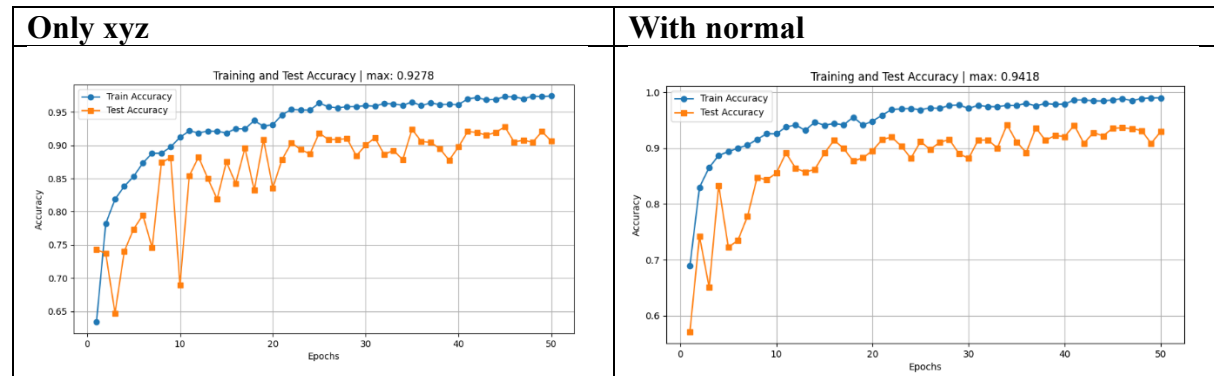
References

- [1] Z. Wu et al., “3D ShapeNets: A Deep Representation for Volumetric Shapes,” arXiv:1406.5670 [cs], Apr. 2015, Available: <https://arxiv.org/abs/1406.5670>
- [2] L. Yi et al., “A scalable active framework for region annotation in 3D shape collections,” ACM Transactions on Graphics, vol. 35, no. 6, pp. 1–12, Nov. 2016, doi: <https://doi.org/10.1145/2980179.2980238>
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- [13] K. L. Cheung and C. C. Lee, “ARCH2S: Dataset, Benchmark and Challenges for Learning Exterior Architectural Structures from Point Clouds,” arXiv.org, 2024. <https://arxiv.org/abs/2406.01337v1>
- [14] Brett, “Master List of 177 Leading 3D Point Cloud Datasets for Industry-Specific Applications,” iMerit, Nov. 28, 2024. <https://imerit.net/blog/master-list-of-177-leading-3d-point-cloud-datasets-for-industry-specific-applications/>

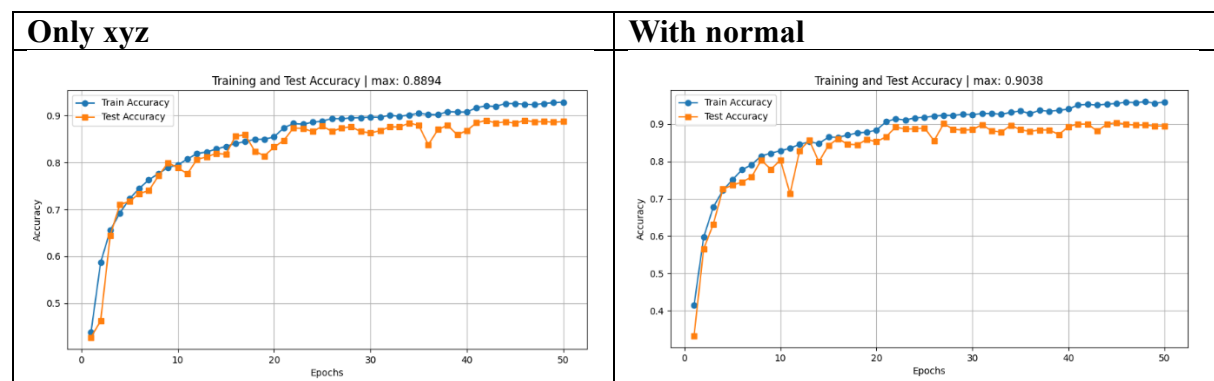
Appendix

(a) PointNet Training

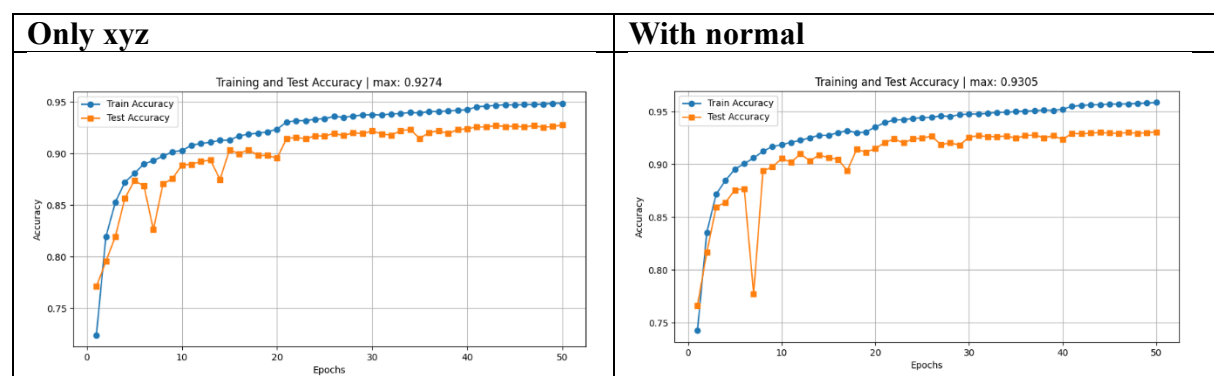
ModelNet10



ModelNet40



ShapeNetCore (part segmentation)

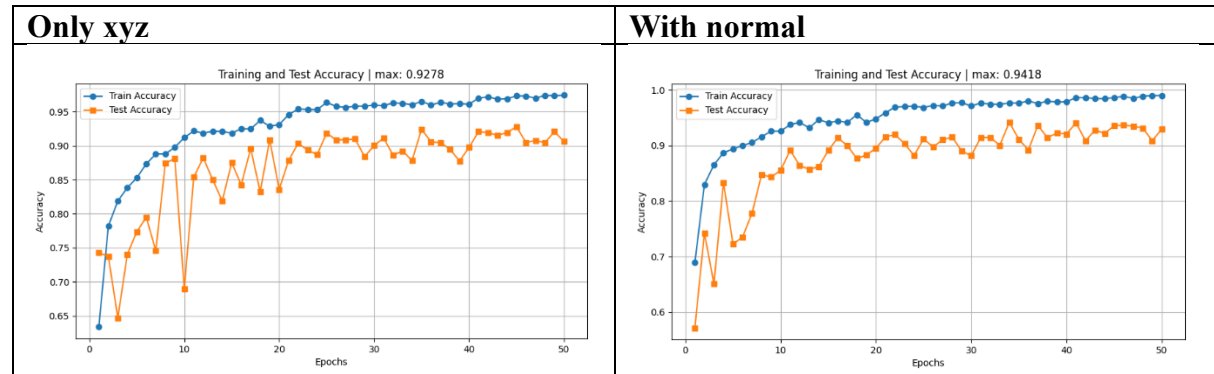


S3DIS (semantic segmentation)

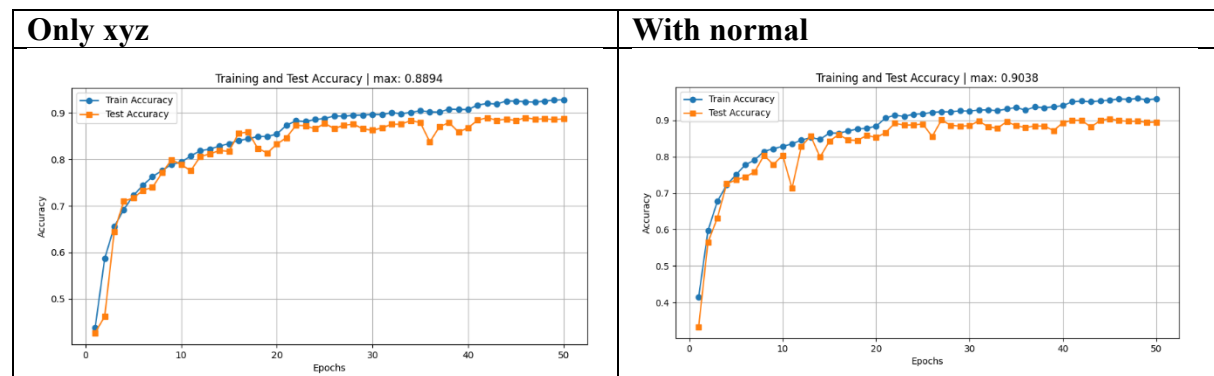
PointNet	https://drive.google.com/drive/folders/1vGtvZFUbaXN_IFfHr0z3z1hiI1XiN8fH?usp=drive_link
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(b) PointNet++ Training

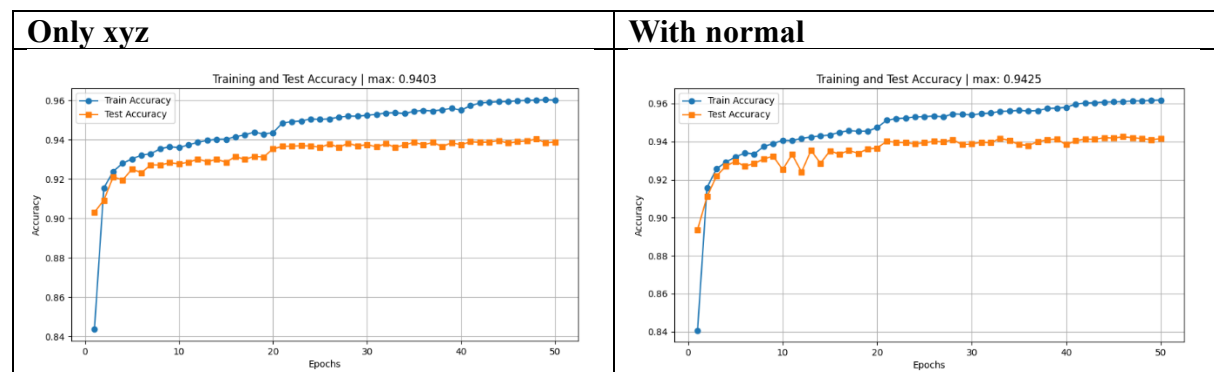
ModelNet10



ModelNet40



ShapeNetCore (part segmentation)



S3DIS (semantic segmentation)

PointNet++	https://drive.google.com/drive/folders/1YHi2GkCNr3d94QG7-JYfkmzA0Lv86POi?usp=drive_link
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(c) Trained model weights

PointNet

ModelNet10 (xyz)	https://drive.google.com/file/d/1vg9PlzLc-8lH8pGVFjkeTe-HQkHiPoZ-/view?usp=sharing
ModelNet10 (xyz + normal)	https://drive.google.com/file/d/1t5fYOC92_6oIh7g_GSvTxkipnipYSn32/view?usp=sharing
ModelNet40 (xyz)	https://drive.google.com/file/d/1qKUTuDdPnP-rQ5ZZMmnUcAMtiPrnqs6u/view?usp=drive_link
ModelNet40 (xyz + normal)	https://drive.google.com/file/d/1ThMo1NaqIn1inWcn8ta m73ZUnUhAx9o_/view?usp=sharing
ShapeNetCore (xyz)	https://drive.google.com/file/d/16JC6scMG_2x12gw5zuE D25k8aeCyXU-O/view?usp=drive_link
ShapeNetCore (xyz + normal)	https://drive.google.com/file/d/1TUFndlLGZ2r7Y0aJ7Q5 PCICOX-9r3tu8/view?usp=sharing
S3DIS	https://drive.google.com/file/d/1wRI3DFu8fkjz1BD2Nm mY8lK-Q1wQorUv/view?usp=drive_link

PointNet++

ModelNet10 (xyz)	https://drive.google.com/file/d/1kDQYkjuE2uCgki_mF3n mnAodQOkli_vl/view?usp=drive_link
ModelNet10 (xyz + normal)	https://drive.google.com/file/d/1yOM0gY00WL9jqB1qc1 RD9-yOt4YL5pLm/view?usp=sharing
ModelNet40 (xyz)	https://drive.google.com/file/d/183Auoop7NOJvsTLi494n dBbirK58nvAC/view?usp=drive_link
ModelNet40 (xyz + normal)	https://drive.google.com/file/d/1LKGVVnu9UORx88Qc-hqgjTFcqr_Hui6u/view?usp=drive_link
ShapeNetCore (xyz)	https://drive.google.com/file/d/1xIasVRGiflhS_NVcsg7FI wmdrVg3qFxt/view?usp=drive_link
ShapeNetCore (xyz + normal)	https://drive.google.com/file/d/1sRr9cN-25yf59KVjBh86ngl9mxAC3IdF/view?usp=drive_link
S3DIS	https://drive.google.com/file/d/1eSd-tu9z9Jq4Jdw-6SROwsjIP6AcNkh3/view?usp=drive_link

(d) Classification Train & Test Logs

PointNet

ModelNet10 (xyz)	https://drive.google.com/file/d/15DMX1f_fMHhaBYdBrL94ObubQoC12eAl/view?usp=drive_link
ModelNet10 (xyz + normal)	https://drive.google.com/file/d/1yBgTmkehNmhid2Dng2hjE17uI9xBREkn/view?usp=drive_link
ModelNet40 (xyz)	https://drive.google.com/file/d/1sKl-C5UJnOyh4YbsF5X2rU5clhVV9u-7/view?usp=drive_link
ModelNet40 (xyz + normal)	https://drive.google.com/file/d/1zNX5Q6RDX2wFmNiPfaW LZdteNrSqOLgN/view?usp=drive_link

PointNet++

ModelNet10 (xyz)	https://drive.google.com/file/d/1-JepNie7XD0abap2MeY0g326YMEOV8PH/view?usp=drive_link
ModelNet10 (xyz + normal)	https://drive.google.com/file/d/12d7M7mFkLFOCzqtBzOLuJRPZw7xIw6Rh/view?usp=drive_link
ModelNet40 (xyz)	https://drive.google.com/file/d/1XVSA28r8O08VljS7JPxrgkAXsEZDxHuh/view?usp=drive_link
ModelNet40 (xyz + normal)	https://drive.google.com/file/d/12fGh8oqgYVo15fjrPFcOEfE1cI6pSnas/view?usp=drive_link

(e) Part Segmentation Evaluation Logs

PointNet (xyz)	https://drive.google.com/file/d/1xHT1e24Pb4nrycT4pSn72PYISMmquVyv/view?usp=drive_link
PointNet (xyz + normal)	https://drive.google.com/file/d/1FqzLcwhti7HZYLdZ_iZnj014CkCeSnLo/view?usp=drive_link
PointNet++ (xyz)	https://drive.google.com/file/d/1njSag69sTdrV46QUhmZOYp0YH1Gh_6QU/view?usp=drive_link
PointNet++ (xyz + normal)	https://drive.google.com/file/d/1Tj27gbAD0VVu01kf1IPNeg5ZXh1v4E7M/view?usp=drive_link

(f) Semantic Segmentation Evaluation Logs

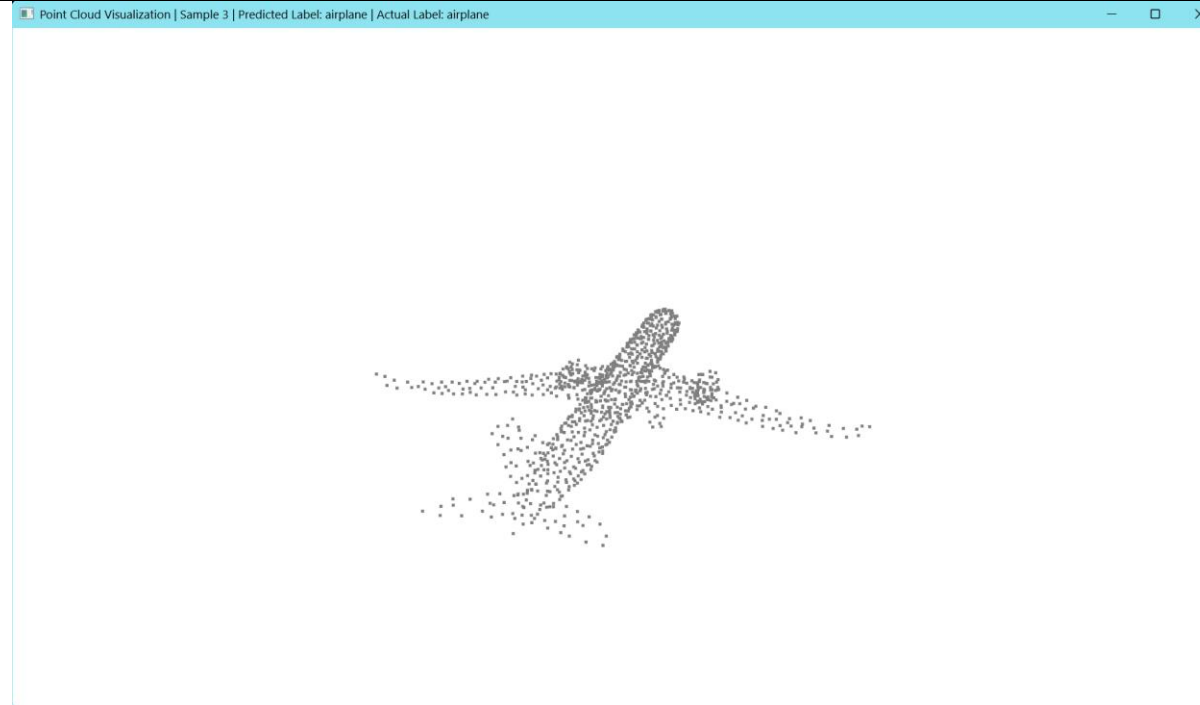
PointNet	https://drive.google.com/file/d/1r5oUeKXBGzogxJnnowhXfUeutC-QnGM3/view?usp=drive_link
PointNet++	https://drive.google.com/file/d/1JKHX2gKQltW23xYVLRECwugIALV6W7A-/view?usp=drive_link

(g) Classification Sample Output

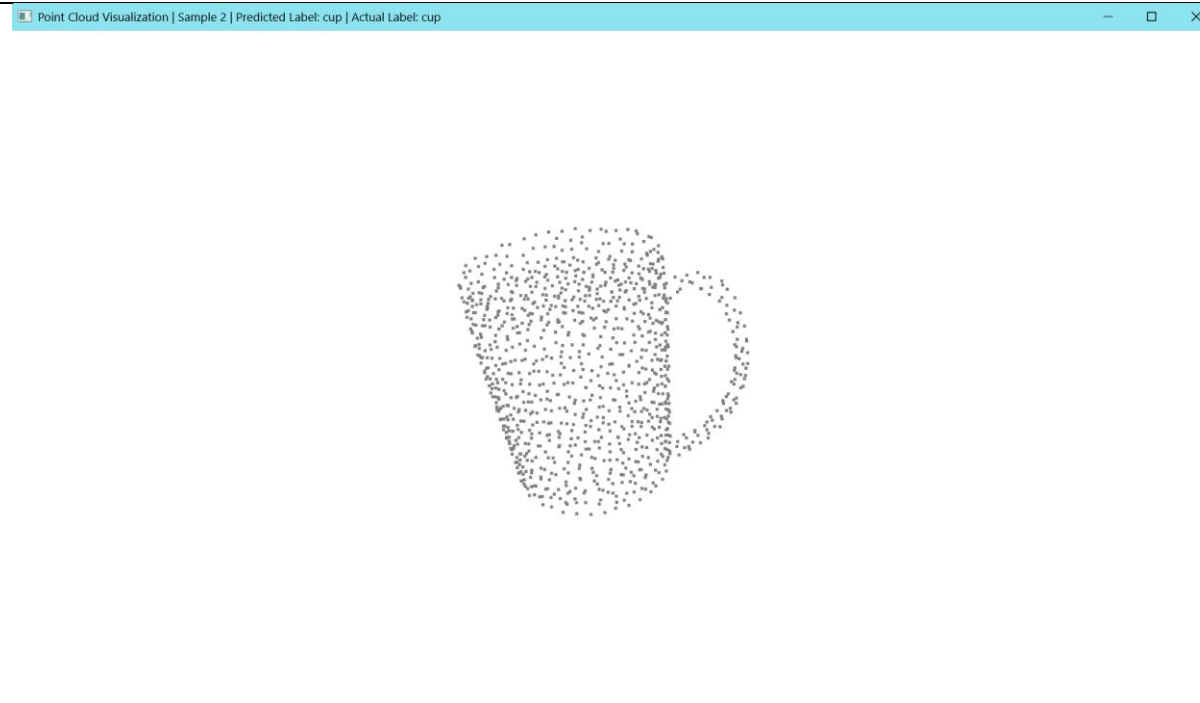
All samples only have xyz coordinates as input.

PointNet

ModelNet40:

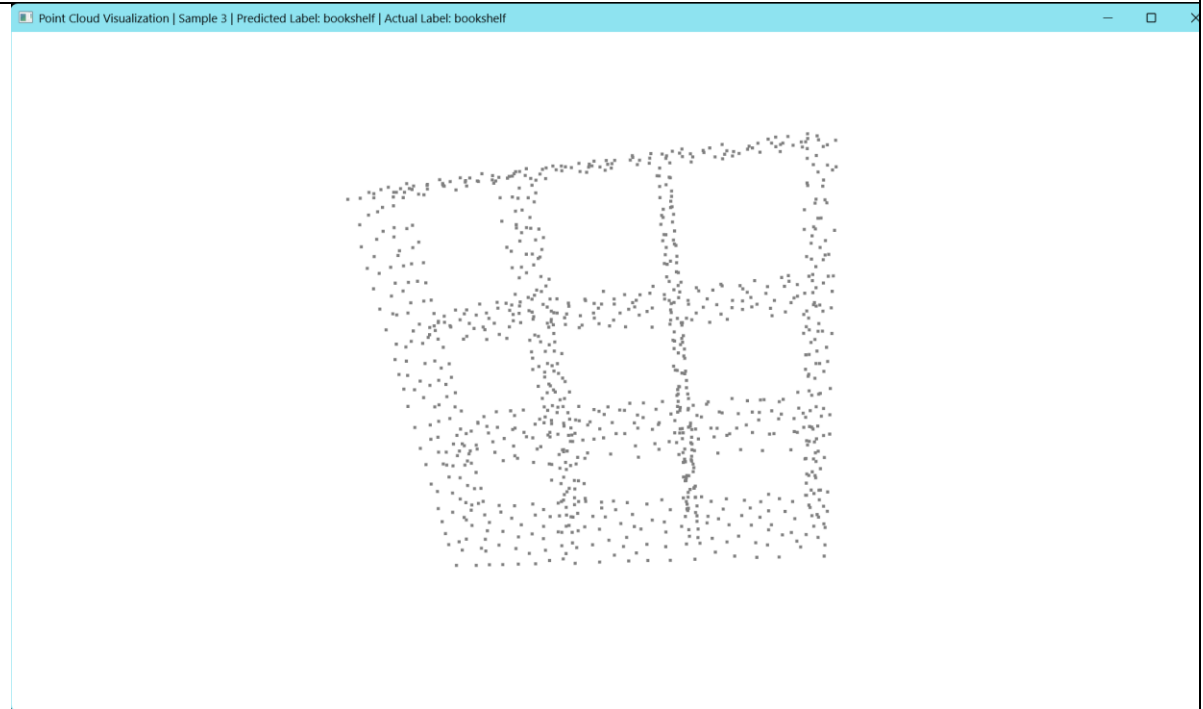


ModelNet40:

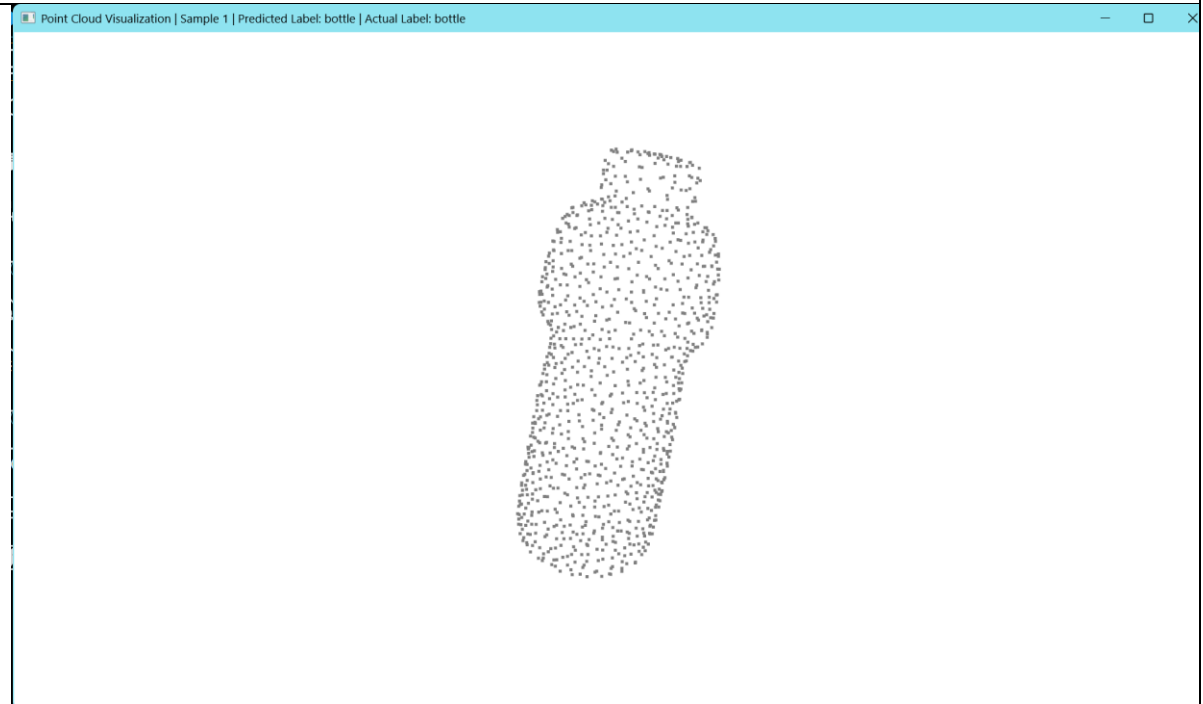


PointNet++

ModelNet40:



ModelNet40:



(h) Part Segmentation Sample Output

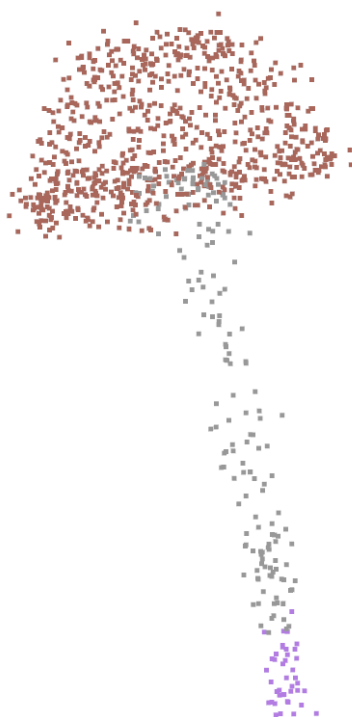
All samples only have xyz coordinates as input.

PointNet

ShapeNetCore:

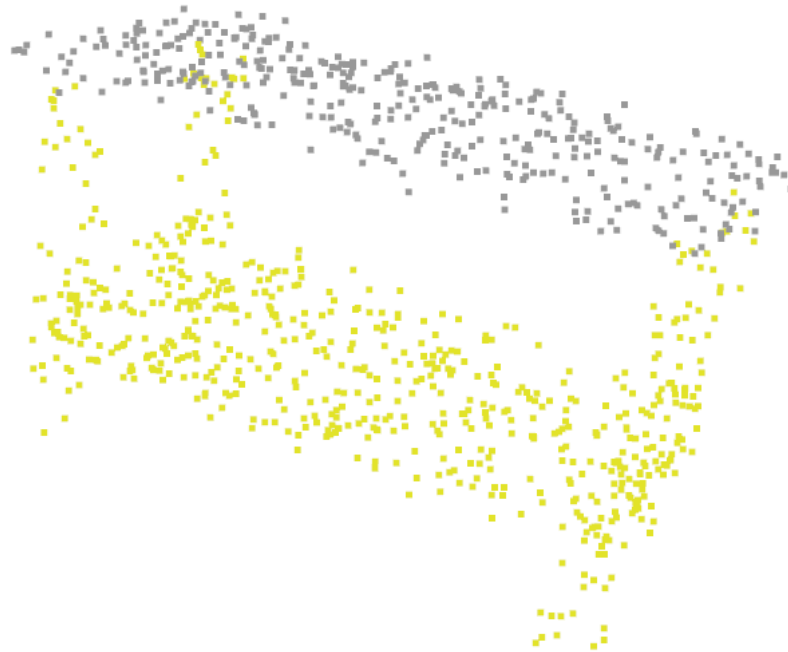


ShapeNetCore:

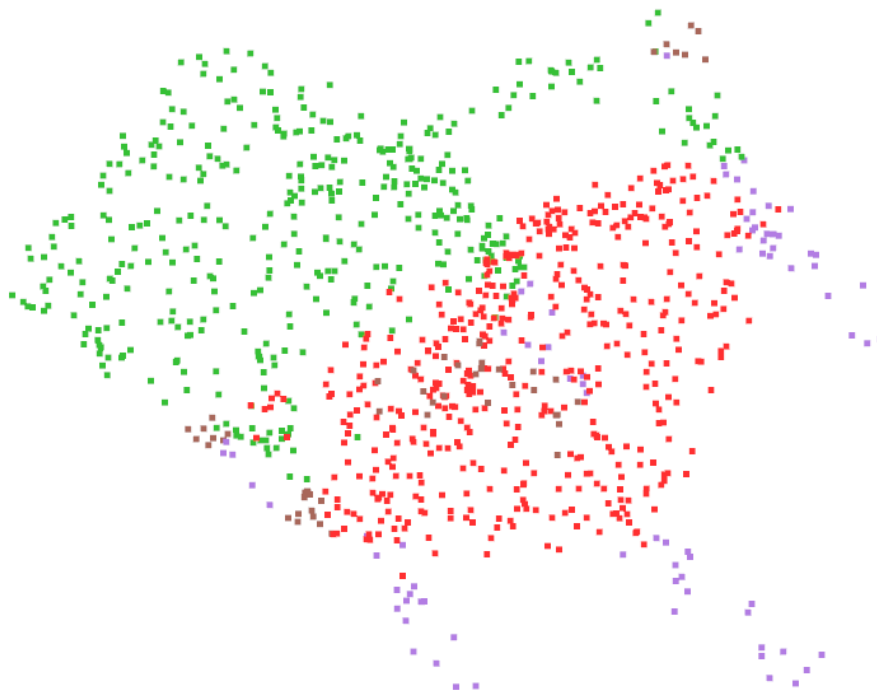


PointNet++

ShapeNetCore:

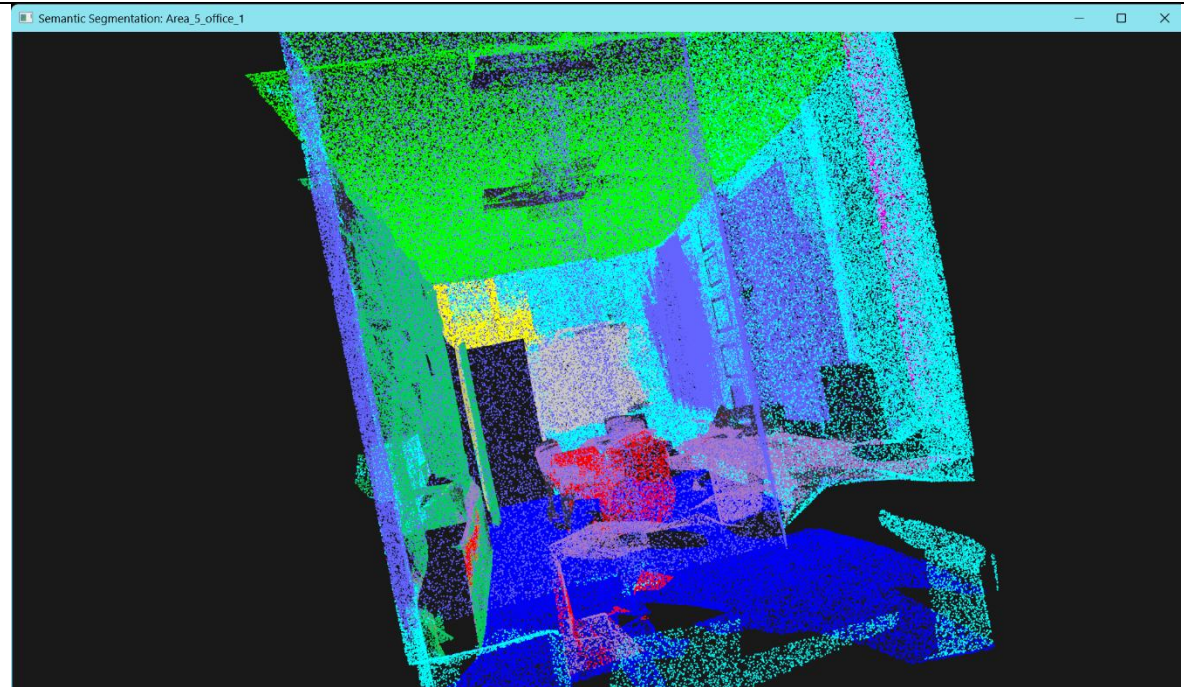


ShapeNetCore:



(i) Semantic Segmentation Sample Output

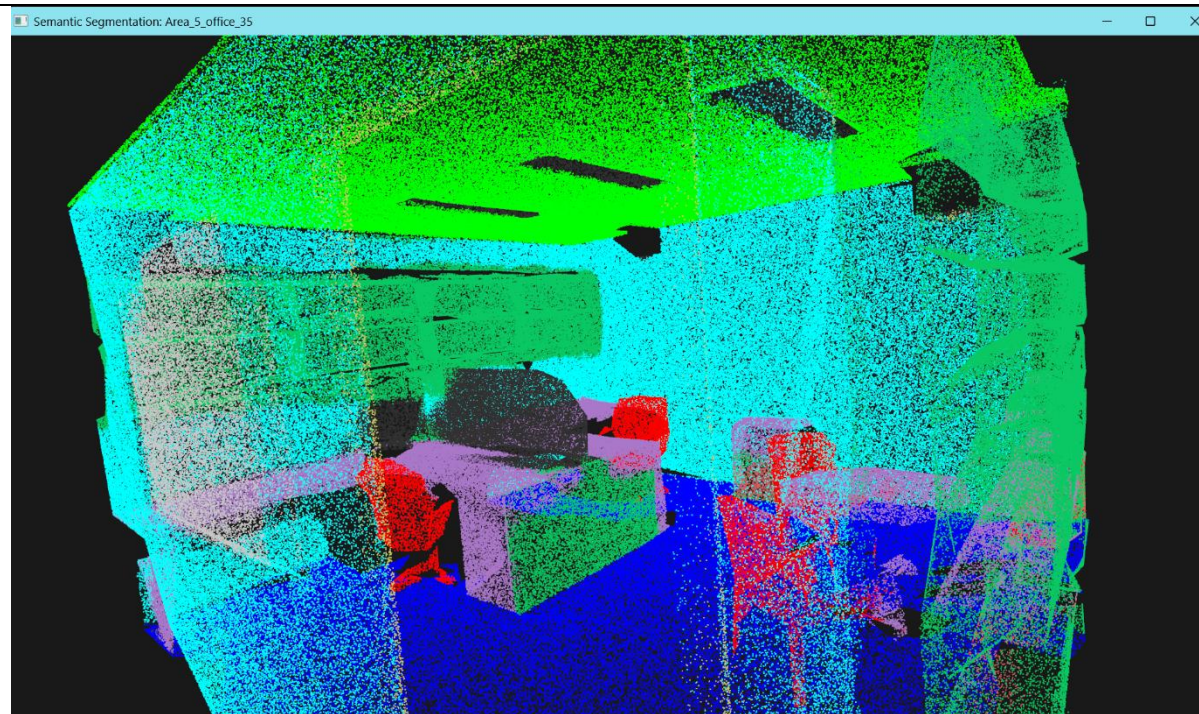
PointNet



All outputs of Area 5 at:

https://drive.google.com/drive/folders/1gsfv1N7rsgwIjSEkd3mB1KZWnUKIYyEG?usp=drive_link

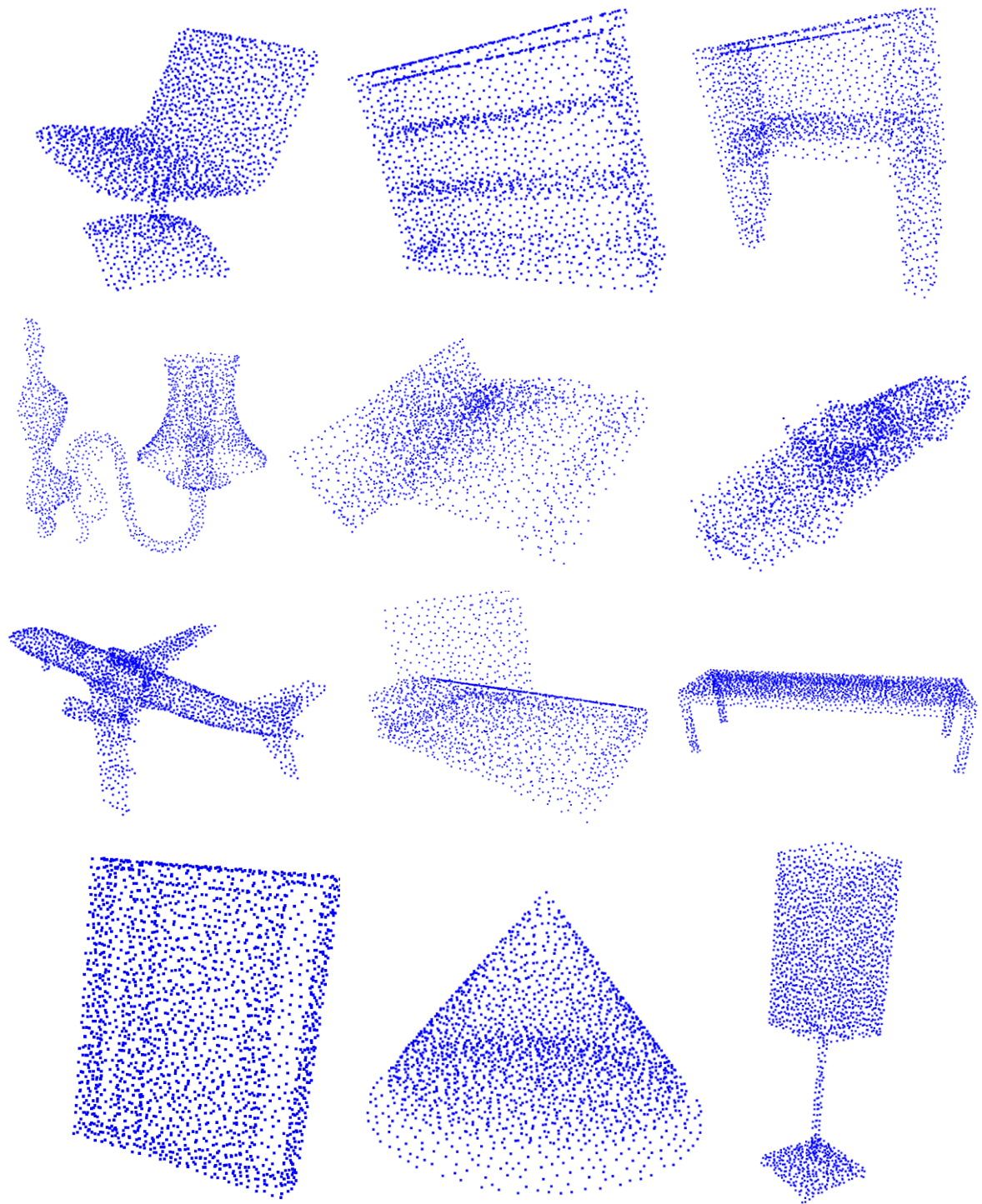
PointNet++



All outputs of Area 5 at: https://drive.google.com/drive/folders/1QWvTUuNznpOljbQ-fF6DU0PuBIYYi9rX?usp=drive_link

(j) More Data Samples

ModelNet40



ShapeNetCore



S3DIS

